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Class : MCA - I.

Subject : Data Structure using Java.

Date : 05-02-2026.

Roll no : 2202561

Type : CCE-01.

pseudo code : Flow chart and dry run.

write pseudo code and draw Flow chart for given problem and explain the pseudo code with proper dry run.

① check given number is palindrome or not.

pseudo code :

Start

Read Number

Set originalNumber = number.

Set reverse = 0.

while, number > 0 DO.

 remainder = number MOD 10.

 reverse = (reverse * 10) + remainder.

 number = number / 10.

End while.

If originalNumber == reverse Then.

 print 'palindrome number'

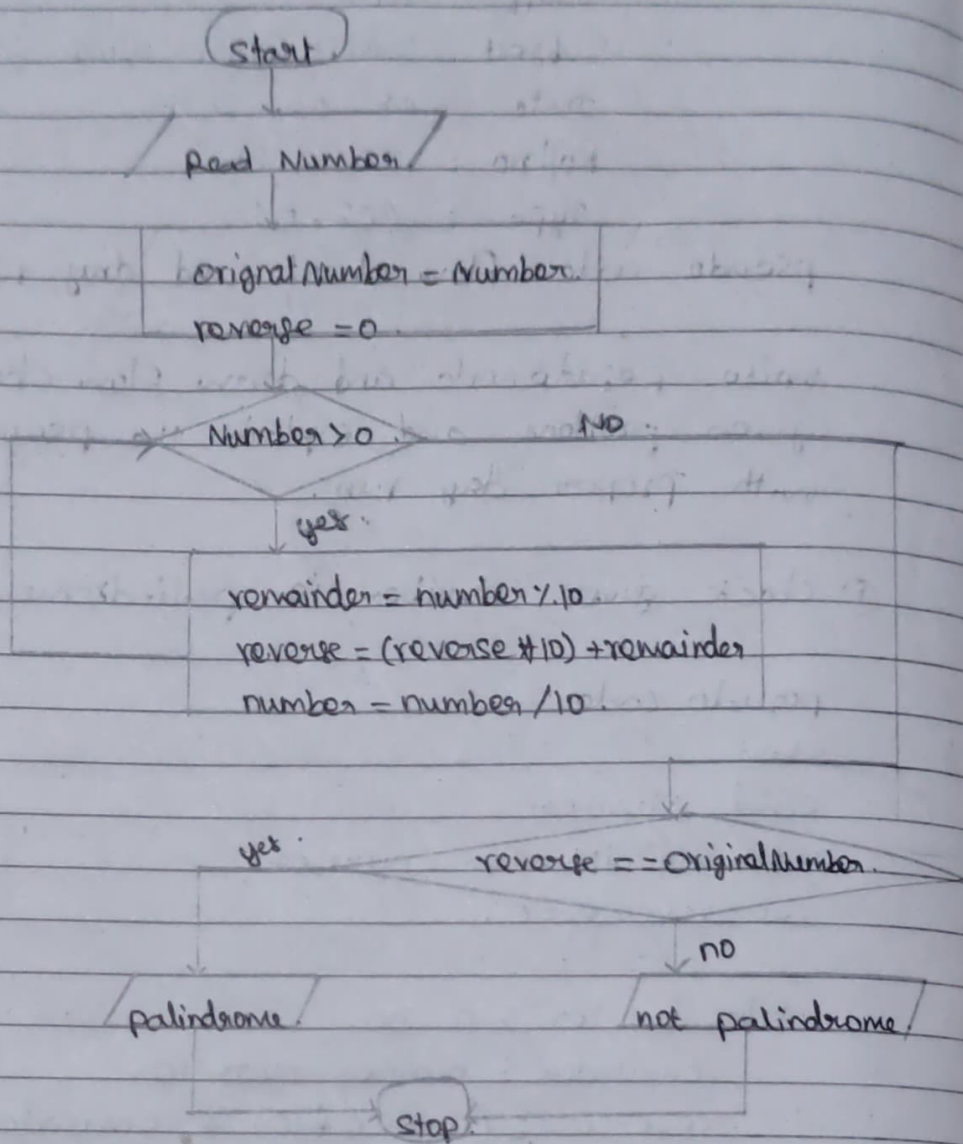
Else

 print 'not palindrome number'

End If

Stop.

Flow chart :



Dry Run :

Start .

Read Number $\longrightarrow$ (181) //

set original Number = Number
(181) $\longleftarrow$ (181).

set reverse = 0 $\longrightarrow$ (0).

number = 181

original Number = 181

reverse = 0 .

(121).
 while Number > 0 Do (✓).
 (1)
 remainder = number % 10.
 1 (101).
 (1) (1) * 10 + 1.
 reverse = (reverse * 10) + remainder
 number = number / 10.
 (12) (121) / (10)
 end while.

①
 remainder = (1).
 reverse = (1)
 number = (12)

(12).
 while Number > 0 Do (✓).
 remainder = number % 10.
 2 (12)
 12 (1) * 10 + 2.
 reverse = (reverse * 10) + remainder
 number = number / 10.
 (1) (12) / (10).

②
 remainder = (2).
 reverse = (12).
 number = (1).

(1).
 while Number > 0 Do (✓).
 remainder = number % 10.
 1 (1)
 reverse = (reverse * 10) + remainder.
 (101). (10) * 10 + 1
 number = number / 10.
 0 1 / 10.

③
 remainder = 1
 reverse = 121
 number = 0.

(0)
 while number > 0 Do (X).
 .
 .
 .
 .
 .

(121) (121).
 If original Number == Reverse Then (✓).
 ↳ print 'palindrome'
 Else
 print 'Not palindrome'

Reverse = 121
 original Number = 121

stop.

② Check. given number is perfect or not.
pseudo code.

Start

Read number.

set $sum = 0$.

For $i = 1$ TO $number / 2$ DO

 If $number \% i == 0$ Then

$sum = sum + i$

 end if

End for

if $sum == number$ Then

 print 'perfect Number'

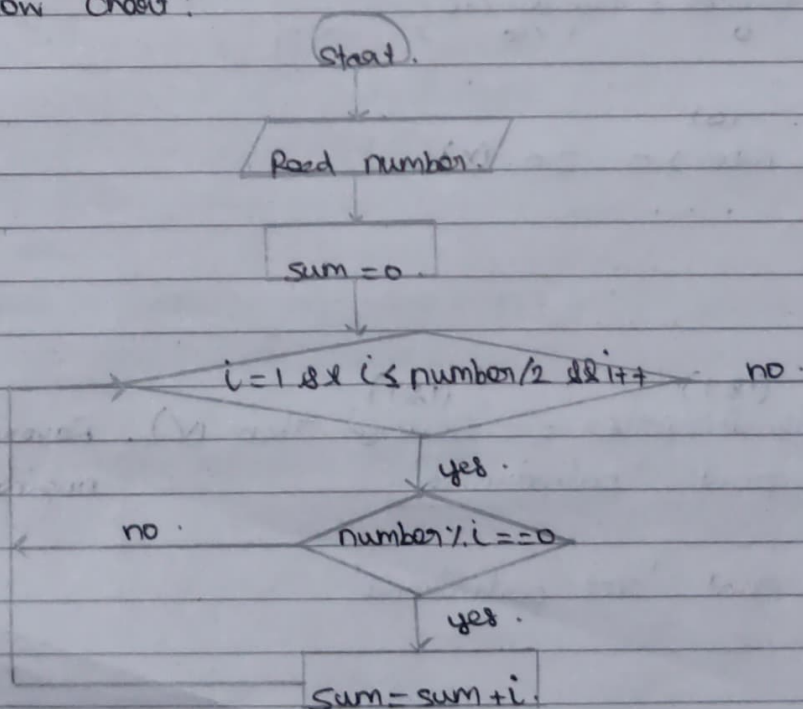
else

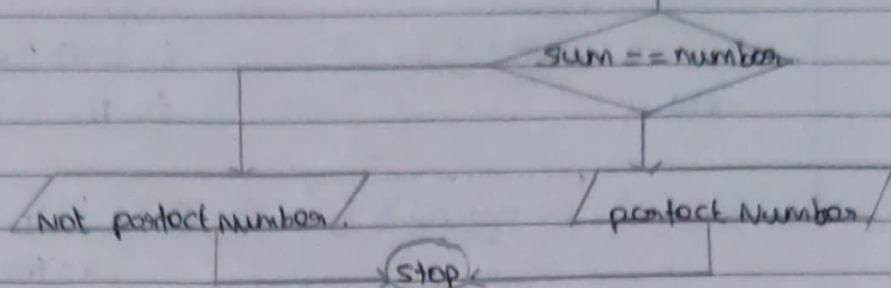
 print 'Not perfect Number'

end if

stop

Flow chart:





Dry run.

Start

Read number $\rightarrow (6)$.

Set sum = 0.

number = (6).

Sum = 0.

for. $i = 1$ & $i \leq \text{number}/2$; $i++$ DO (✓).

①.

$i = 1$ & $1 \leq 6/2$; $1++$
(3).

$i = 1$

$1 \leq 6/2 \Rightarrow 1 \leq 3 \checkmark$

if. number % $i == 0$ Then (✓).

$6 \% 1 == 0 \checkmark$

$6 \% 1 == 0$

Sum = $0 + 1 \Rightarrow 1$

Sum = Sum + i

$0 + 1 = 1$

$i++ \Rightarrow 1++ \Rightarrow 2$

end if.

$i = 2$

for. $i = 2$ & $i \leq \text{number}/2$; $i++$ DO (✓).

②.

$i = 2$ & $2 \leq (6/2)$; $2++$
(3).

$i = 2$

$2 \leq 6/2 \Rightarrow 2 \leq 3 \checkmark$

if. number % $i == 0$ Then. (✓).

$6 \% 2 == 0 \checkmark$

$6 \% 2 == 0$

Sum = $1 + 2 \Rightarrow 3$

Sum = Sum + i

$3 = 1 + 2$

$i++ \Rightarrow 2++ \Rightarrow 3$

end if.

$i = 3$

for. $i = 3$ & $i \leq \text{number}/2$ & $i++$ DO (✓).

③.

$i = 3$ & $3 \leq (6/2)$ & $3++$
(3)

$i = 3$

$3 \leq 6/2 \Rightarrow 3 \leq 3 \checkmark$

if. number % $i == 0$ Then. (✓).

$6 \% 3 == 0 \checkmark$

$6 \% 3 == 0$

Sum = $3 + 3 \Rightarrow 6$

sum = sum + i
6 = 3 + 3
end it.

i++ $\Rightarrow$ 3++ $\Rightarrow$ 4
i = 4.

for i = 4 $\&\&$ i < number/2 $\&\&$ it do (X)

i = 4
 $4 \leq 6/2 \Rightarrow 4 \leq 3$ (X)

(6) (6)
it. sum == number. Then
 print 'perfect'
else
 print 'Not perfect'
endit
stop.

sum = 6
number = 6.

③ write a pseudo code to insert element in between array.

pseudo code:

Start.

Read n

Declare Array A of size n+1

For i = 0 to n-1 DO

 Read A[i]

End For.

Read element.

Read position.

For i = n down to position DO

 A[i] = A[i-1]

End For.

A[position] = element.

n = n+1

Print 'Array after insertion'

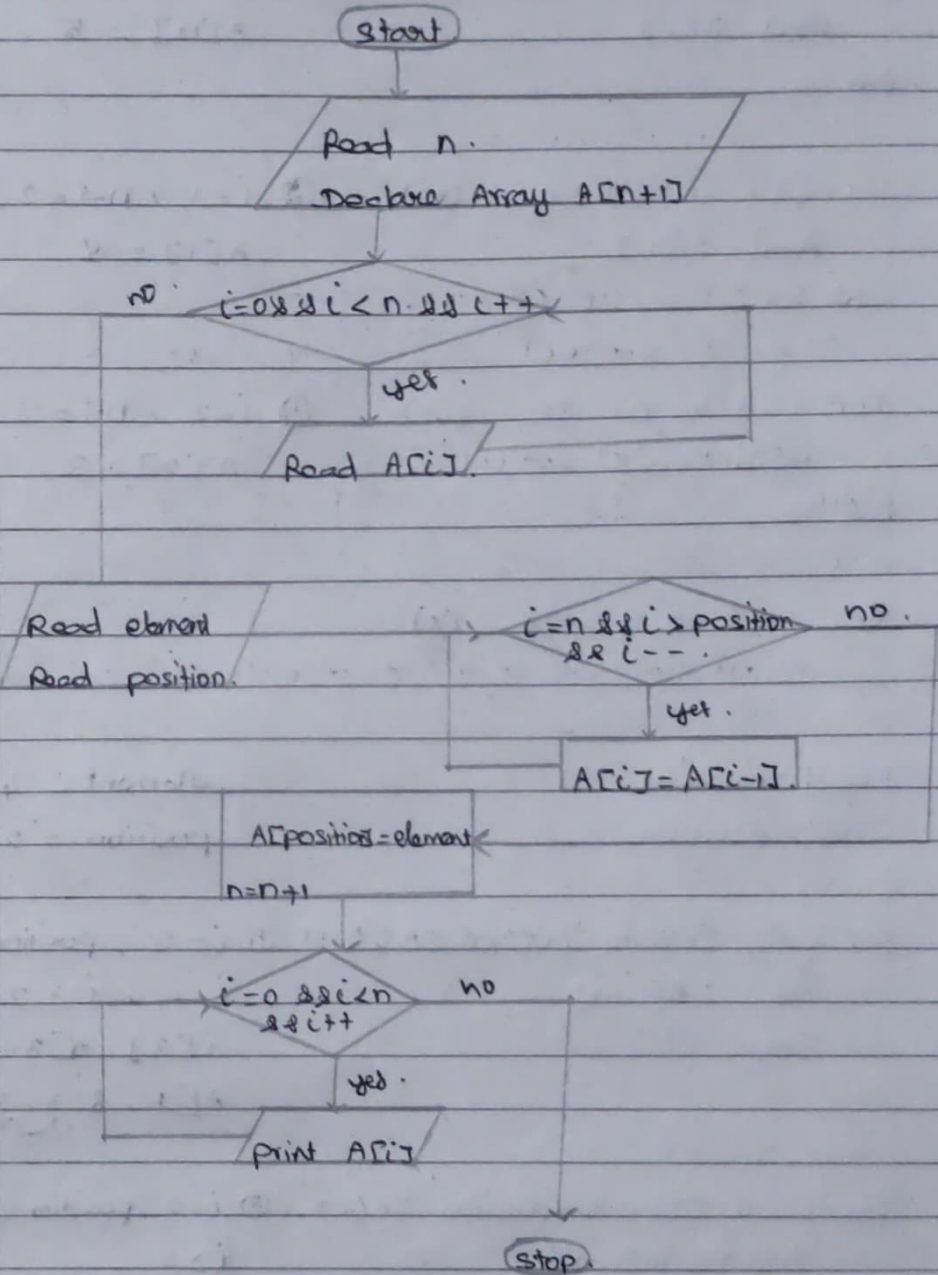
for $i=0$ to $n-1$ DO

print $A[i]$

end for

Stop

Flow chart:



Dry run:

Start.

read $n \rightarrow (3)$.

$n = 3$

declare Array A of Size $n+1$.

$A[n+1] \Rightarrow A[3+1] \Rightarrow A[4]$

For $i=0$ to $n-1$ DO $(\checkmark)$ ①

$i=0 ; n-1=2 ; 0 \leq 2$

Read $A[i]$

$A[0] = 5$

End For.

For $i=1$ to $n-1$ DO $(\checkmark)$ ②

$i=1 ; n-1=2 ; 1 \leq 2$

Read $A[i]$

$A[1] = 8$

End For.

For $i=2$ to $n-1$ DO $(\checkmark)$ ③

$i=2 ; n-1=2 ; 2 \leq 2$

Read $A[i]$

$A[2] = 3$

End For.

For $i=3$ to $n-1$ DO (X)

.....

Read element

element = 4

Read position.

position = 2

for $i=3$ Downto position DO (X) ①

$i=3 ; position=2 ;$

$A[i] = A[i-1]$

$i-- ; i \geq 2$

End For.

$A[3] = A[3-1]$

$A[5, 8, 3, 3]$

for $i=2$ Downto position DO (X) ②

$i=2 ; position=2 ; i-- ;$

$A[i] = A[i-1]$

$i \geq 2$

End For.

$A[2] = A[2-1]$

$A[5, 8, 8, 3]$

For $i=1$ Downto position DO (X) ③

$i \geq 2$ (X).

.....

position j + element

$n = n + 1$

$A[2] = 4$

$n = 3 + 1 \Rightarrow n = 4$

$A[5, 8, 4, 3]$
0 1 2 3

print 'Alter insertion'

for $i = 0$ to $n - 1$ DO (✓) ①
 $i = 0$ 4-1

$i = 0 ; 0 < 4 ; i++$

$A[0] \Rightarrow 5$

print $A[i]$
0

end for.

for $i = 1$ to $n - 1$ DO (✓) ②
 $i = 1$ 4-1

$i = 1 ; 1 < 4 ; i++$

$A[i] \Rightarrow 8$

print $A[i]$
1

end for.

for $i = 2$ to $n - 1$ DO (✓) ③
 $i = 2$ 4-1

$i = 2 ; 2 < 4 ; i++$

$A[2] \Rightarrow 2$

print $A[i]$
2

end for.

for $i = 3$ to $n - 1$ DO (✓) ④
 $i = 3$ 4-1

$i = 3 ; 3 < 4 ; i++$

$A[3] \Rightarrow 3$

print $A[i]$
3

end for.

for $i = 4$ to $n - 1$ DO (X)
 $i = 4$ 4-1

...

...

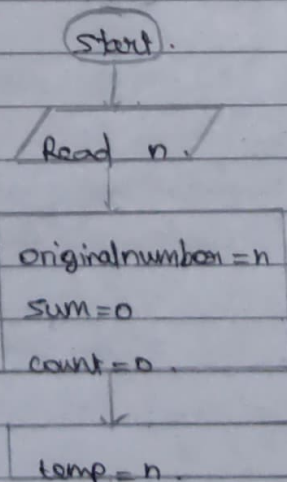
stop.

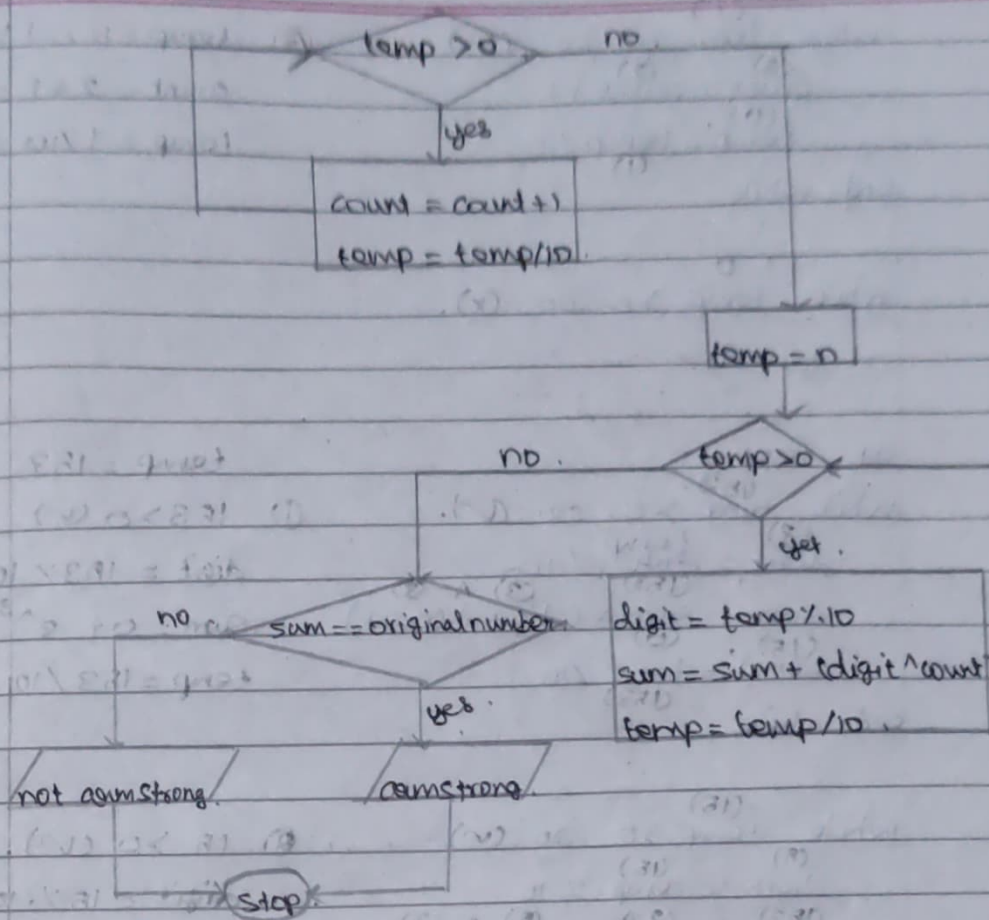
④ Check whether given number is armstrong or not
pseudocode:

```

Start
Read n
set originalnumber = n
set sum = 0
set count = 0
set temp = number
while temp > 0 do
    count = count + 1
    temp = temp / 10
end while
set temp = number
while temp > 0 do
    digit = temp % 10
    sum = sum + (digit ^ count)
    temp = temp / 10
end while
if sum == originalnumber then
    print 'armstrong'
else
    print 'not armstrong'
end if
Stop
    
```

flow chart:





Dry run:

Start

read $n \rightarrow (153)$

set original number = n

set sum = 0

set count = 0

set temp = n

while temp > 0 do

(1) count = count + 1

(15) temp = temp / 10

end while

(15) while temp > 0 do

(2) (1) count = count + 1

(1) temp = temp / 10

end while

$n = 153$

original number = 153

sum = 0

count = 0

temp = 153

①. $153 > 0 (\checkmark)$

count = $0 + 1 \Rightarrow 1$

temp = $153 / 10 \Rightarrow 15$

temp = 15

②. temp = 15 ; $15 > 0 (\checkmark)$

count = $1 + 1 \Rightarrow 2$

temp = $15 / 10 \Rightarrow 1$

(1) while temp > 0 DO (✓) ③ temp = 1 ; 1 > 0 (✓)
 (3) count = count + 1
 (0) temp = temp / 10
 (1) end while

count = 2 + 1 ⇒ 3 //
 temp = 1 / 10 ⇒ 0

while temp > 0 DO (X).

set temp = n.
 while temp > 0 DO (✓).
 (3) digit = temp % 10
 (153) (3) ^ (3)
 sum = sum + (digit ^ count)
 (15) (0)
 temp = temp / 10
 (153)
 end while

temp = 153.
 ① 153 > 0 (✓).
 digit = 153 % 10 ⇒ 3.
 sum = 0 + 3³ ⇒ 27
 temp = 153 / 10 ⇒ 15

(15) while temp > 0 DO (✓).
 (5) (15)
 digit = temp % 10
 (152) (27) (5) ^ (3)
 sum = sum + (digit ^ count)
 (11)
 temp = temp / 10
 (15)
 end while

② 15 > 0 (✓).
 digit = 15 % 10 ⇒ 5
 sum = 27 + 5³ ⇒ 152
 temp = 15 / 10 ⇒ 1

(1) while temp > 0 DO (✓)
 (1) (1)
 digit = temp % 10
 (153) (152) (1) ^ (3)
 sum = sum + (digit ^ count)
 (0)
 temp = temp / 10
 (1)
 end while

③ 1 > 0 (✓).
 digit = 1 % 10 ⇒ 1
 sum = 152 + 1³ ⇒ 153
 temp = 1 / 10 ⇒ 0.

while temp > 0 DO (X).

if sum == original number then (✓).

print 'armstrong'

else

print 'not armstrong'

end if

stop.

⑤ Accept array elements and find maximum and minimum elements from the array.

pseudo code:

Start

Read n.

declare Array A of Size n.

for $i=0$ TO $n-1$ DO.

Read $A[i]$.

End for.

set $MIN=0$.

set $MAX=0$.

for $i=0$ TO $n-1$ DO.

if $A[i] > MAX$ Then.

$MAX = A[i]$

End if.

if $A[i] < MIN$ Then.

$MIN = A[i]$.

End if.

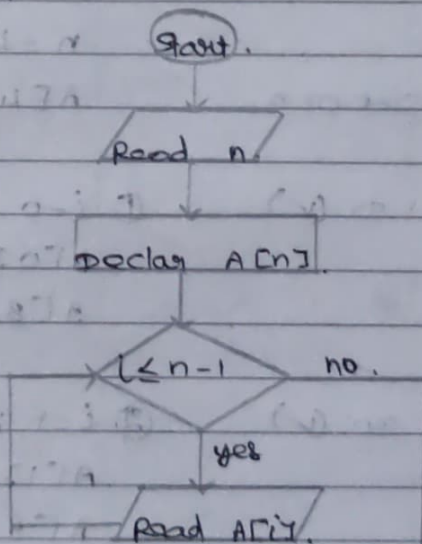
End for.

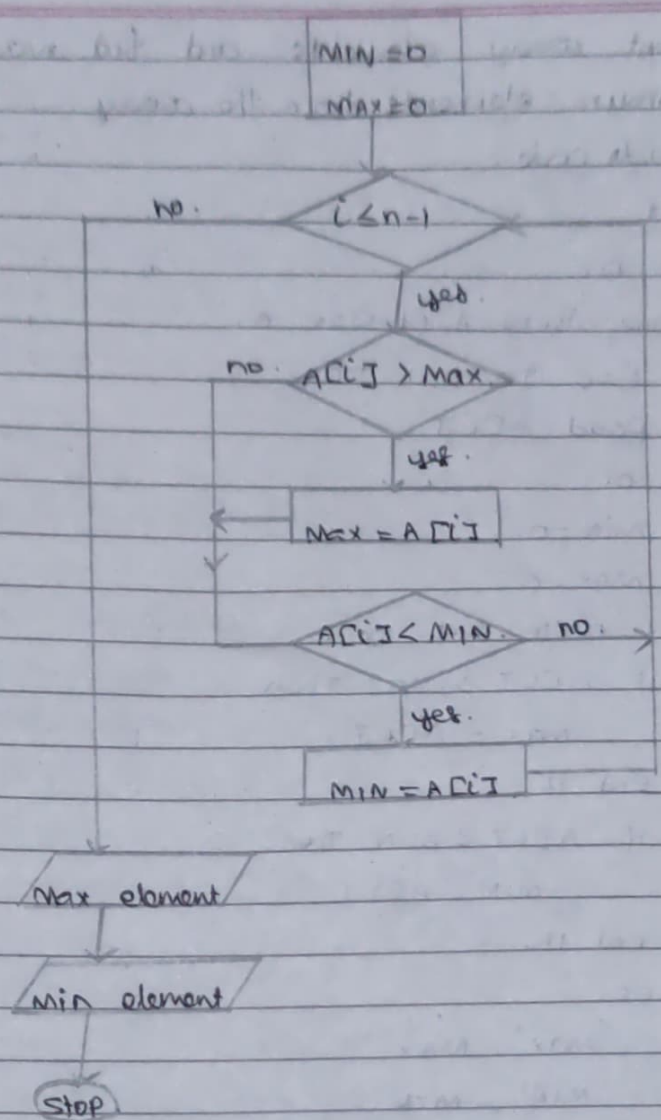
print 'MAX', MAX

print 'MIN', MIN.

Stop.

flow chart.





Day Run:

Start

Read n (4)

n = 4 0 1 2 3
A[4] = [0, 0, 0, 0]

Declare Array A size of n.

for i = 0 to n-1 do (✓)

① i = 0 ; 0 ≤ 4-1 (✓)

 Read A[i]
 (0)

A[0] = 8.

end for.

A[8, 0, 0, 0].

for i = 1 to n-1 do (✓)

② i = 1 ; 1 ≤ 4-1 (✓)

 Read A[i]
 (1)

A[1] = 1

end for.

A[8, 1, 0, 0].

for: $i=2$ TO $n-1$ DO (✓) ③	$i=2 ; 2 \leq 4-1$ (✓)
Read $A[i]$	$A[2] = 4$
end For	$A[8, 1, 4, 0]$

for $i=2$ TO $n-1$ DO (✓) ④	$i=3 ; 3 \leq 3$ (✓)
Read $A[i]$	$A[3] = 7$
end For	$A[8, 1, 4, 7] //$

$4 \leq 3$

For: $i=4$ TO $n-1$ DO (x)

For: $i=0$ TO $n-1$ DO (✓) ①	$i=0 ; 0 \leq 3$ (✓)
If $A[i] < \text{max}$ Then	$8 < 0$ (✓)
$\text{max} = A[i]$	$\text{max} = 8$
End If	
If $A[i] > \text{min}$ Then	$8 > \phi$ (x)
$\text{min} = A[i]$	$\text{min} = \phi$
end If	
end For	

For: $i=1$ TO $n-1$ DO (✓) ②	$i=1 ; 1 \leq 3$ (✓)
If $A[i] < \text{max}$ Then	$1 < 8$ (x)
$\text{max} = A[i]$	$\text{max} = 8$
end If	
If $A[i] > \text{min}$ Then	$1 > 1$ (x)
$\text{min} = A[i]$	$\text{min} = 8$
end If	
end For	

for: $i=2$ TO $n-1$ DO (✓) ③	$i=2 ; 2 \leq 3$ (✓)
If $A[i] < \text{max}$ Then	$4 > 8$ (x)
$\text{max} = A[i]$	$\text{max} = 8$
end If	
If $A[i] > \text{min}$ Then	$4 < 1$ (x)
$\text{min} = A[i]$	$\text{min} = 1$
end for end if	

For $i = 2$ to $n-1$ DO (✓)
 IF $A[i] < \text{max}$ THEN
 $\text{max} = A[i]$
 and if.
 IF $A[i] > \text{min}$ THEN
 $\text{min} = A[i]$
 and if.
 and for.

(4) $i = 3 ; 2 \leq 3$ (✓)
 $7 > 8$ (✓)
 $\text{max} = 8 //$
 $7 < 1$ (✓)
 $\text{min} = 1 //$

$4 \geq 3$
 FOR $i = 4$ TO $n-1$ DO (X).

print 'max', max
 print 'min', min
 stop